# AiCo

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Software engineering Assignment 2
BCS 6J

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# 1. Functional Requirements

The following are the 10 main functional requirements of the AiCo system:

1. **User Registration and Authentication:** Allows instructors and students to create an account and securely log in to access their role-specific dashboard.
2. **Classroom Creation and Management:** Enables instructors to create virtual classrooms, configure settings, and manage enrolled students.
3. **Student Enrollment in Classrooms:** Allows students to join classrooms using a unique code provided by the instructor.
4. **Learning Material Upload and Management:** Enables instructors to upload, organize, and manage learning materials such as PDFs and documents within each classroom.
5. **AI-Powered Learning Assistant (RAG-Based):** Provides students with a context-aware AI assistant that answers questions strictly grounded in instructor-uploaded content using RAG.
6. **Content Summarization:** Allows students to request AI-generated summaries of uploaded classroom materials to aid in revision and comprehension.
7. **Automated Marksheet Generation:** Enables instructors to automatically compile student grades from quizzes and assignments and export them as a formatted Google Sheets marksheet.
8. **Assignment and Quiz Submission with Plagiarism Detection:** Allows students to submit assignments and quizzes; the system automatically analyzes submissions for text similarity and generates plagiarism reports.
9. **Interactive Student Study Whiteboard:** Provides students with a digital whiteboard within each classroom where they can write notes, draw diagrams, and organize study content visually.
10. **Pomodoro Timer for Study Management:** Provides students with a built-in Pomodoro timer to manage focused study sessions and breaks, operating independently from AI services.

# 2. Use Case Descriptions

## UC-01 – User Registration and Authentication

| Identifier | UC-01 |
|---|---|
| **Name** | User Registration and Authentication |
| **Summary** | Allows instructors and students to create an account and securely log in to access their role-specific dashboard. |
| **Priority** | High |
| **Actors** | Instructor, Student, System (auth service) |
| **Pre-condition(s)** | User does not have an existing account. |
| **Post-condition(s)** | A new user account is created and the user is authenticated. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| User navigates to AiCo and clicks Sign Up. | Displays the registration form. |
| User enters name, email, password, and selects role (Instructor/Student). | Captures submitted data and waits for submission. |
| User clicks Create Account. | Validates format and uniqueness of email. |
| | Creates account and redirects user to their dashboard. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| User enters an already registered email. | Displays 'Email already exists' error and prompts use of a different email. |
| User submits with weak password or empty fields. | Displays validation error messages and prompts correction. |

## UC-02 – Classroom Creation and Management

| Identifier | UC-02 |
|---|---|
| **Name** | Classroom Creation and Management |
| **Summary** | Enables instructors to create virtual classrooms, configure settings, and manage enrolled students. |
| **Priority** | High |
| **Actors** | Instructor, System |
| **Pre-condition(s)** | Instructor is logged in. |
| **Post-condition(s)** | Classroom is created and visible on the instructor's dashboard. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor clicks 'Create Classroom' from dashboard. | Displays classroom creation form. |
| Instructor enters classroom name, subject, and description. | Captures input and waits for submission. |
| Instructor submits the form. | Generates a unique classroom code and stores classroom data. |
| | Redirects instructor to the new classroom's management page. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor submits with an empty required field. | Displays validation error and prevents submission. |
| Instructor tries to use a duplicate classroom name. | Warns instructor and prompts entry of a different name. |

# UC-03 – Student Enrollment in Classrooms

| Identifier | UC-03 |
|---|---|
| **Name** | Student Enrollment in Classrooms |
| **Summary** | Allows students to join classrooms using a unique code provided by the instructor. |
| **Priority** | High |
| **Actors** | Student, System |
| **Pre-condition(s)** | Student is logged in and holds a valid classroom code. |
| **Post-condition(s)** | Student is enrolled and can access all materials in the classroom. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student clicks 'Join Classroom' from dashboard. | Displays the classroom code entry form. |
| Student enters the unique classroom code. | Validates the code against existing classrooms. |
| | Adds student to the classroom roster. |
| | Redirects student to the joined classroom page. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student enters an invalid or expired code. | Displays 'Invalid code' error message. |
| Student attempts to join a classroom they are already enrolled in. | Displays 'Already enrolled' message and prevents duplicate enrollment. |

## UC-04 – Learning Material Upload and Management

| Identifier | UC-04 |
|---|---|
| **Name** | Learning Material Upload and Management |
| **Summary** | Enables instructors to upload, organize, and manage learning materials such as PDFs and documents within each classroom. |
| **Priority** | High |
| **Actors** | Instructor, System |
| **Pre-condition(s)** | Instructor is logged in and has at least one classroom. |
| **Post-condition(s)** | Materials are stored, indexed for AI retrieval, and accessible to enrolled students. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor opens a classroom and clicks 'Upload Material'. | Displays file selection dialog. |
| Instructor selects a file from their device. | Validates file type and size. |
| | Stores the file and indexes it for AI-based retrieval. |
| | Displays uploaded material in the classroom's material library. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor uploads an unsupported file format. | Rejects the upload and notifies instructor of supported formats. |
| File exceeds the size limit. | Displays size limit error and prevents upload. |

## UC-05 – AI-Powered Learning Assistant (RAG-Based)

| Identifier | UC-05 |
|---|---|
| **Name** | AI-Powered Learning Assistant (RAG-Based) |
| **Summary** | Provides students with a context-aware AI assistant that answers questions strictly grounded in instructor-uploaded content using RAG. |
| **Priority** | High |
| **Actors** | Student, System (AI service) |
| **Pre-condition(s)** | Student is enrolled in a classroom with uploaded materials. |
| **Post-condition(s)** | Student receives an accurate, material-grounded answer to their question. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|

| | |
|---|---|
| Student opens the AI assistant within a classroom. | Displays the assistant chat interface. |
| Student types a question related to course material. | Retrieves relevant content chunks from indexed materials. |
| | Generates a context-aware response grounded in retrieved content. |
| | Displays the response with relevant references to the student. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Query is outside the scope of uploaded materials. | Informs student that the query is beyond available material. |
| AI service is temporarily unavailable. | Displays an error message and suggests trying again later. |

## UC-06 – Content Summarization

| Identifier | UC-06 |
|---|---|
| **Name** | Content Summarization |
| **Summary** | Allows students to request AI-generated summaries of uploaded classroom materials to aid in revision and comprehension. |
| **Priority** | Medium |
| **Actors** | Student, System (AI service) |
| **Pre-condition(s)** | Student is enrolled and materials are available in the classroom. |
| **Post-condition(s)** | A readable AI-generated summary of the material is presented to the student. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student opens a classroom and selects a document. | Displays document details and options. |
| Student clicks 'Summarize'. | Retrieves document content and sends it to the AI summarization service. |
| | AI generates a concise summary of the key concepts. |
| | Displays the summary to the student. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Document is too large to process at once. | System processes it in segments and combines the summaries. |
| AI service fails to generate a summary. | Displays an error and prompts the student to retry. |

## UC-07 – Automated Marksheet Generation

| Identifier | UC-07 |
|---|---|
| **Name** | Automated Marksheet Generation |
| **Summary** | Enables instructors to automatically compile student grades from quizzes and assignments and export them as a formatted Google Sheets marksheet |
| **Priority** | High |
| **Actors** | Instructor, System (Google Sheets API) |
| **Pre-condition(s)** | Grades have been entered or finalized for at least one assessment in the classroom |
| **Post-condition(s)** | A formatted Google Sheets marksheet is generated and accessible to the instructor |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor navigates to the grading section of a classroom. | Displays all assessments and current grade entries. |
| Instructor clicks 'Export Marksheet'. | Compiles all student grades across assessments into a structured format. |
| | |
| | |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor attempts to export with incomplete grades. | Displays a warning listing students with missing grades and prevents export until resolved.. |
| Google Sheets API is unavailable. | Displays an error message and offers to export as a local CSV file instead. |

## UC-08 – Assignment and Quiz Submission with Plagiarism Detection

| Identifier | UC-08 |
|---|---|
| **Name** | Assignment and Quiz Submission with Plagiarism Detection |
| **Summary** | Allows students to submit assignments and quizzes; the system automatically analyzes submissions for text similarity and generates plagiarism reports. |
| **Priority** | High |
| **Actors** | Student, Instructor, System (similarity engine) |

| | |
|---|---|
| **Pre-condition(s)** | An assignment or quiz has been created by the instructor. |
| **Post-condition(s)** | Submission is stored, plagiarism report is generated, and the instructor is notified. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Instructor creates an assignment with a deadline. | Stores assignment and makes it visible to enrolled students. |
| Student writes and submits their response. | Receives submission and triggers the text similarity analysis. |
| | Calculates similarity scores against prior submissions and reference material. |
| | Generates a plagiarism report and notifies the instructor. |
| Instructor reviews the report. | Final grading decision remains with the instructor. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student submits after the deadline. | Flags submission as late and notifies the instructor. |
| Similarity engine fails. | Logs the error and notifies instructor that the report is pending. |

# UC-09 – Interactive Student Study Whiteboard

| Identifier | UC-09 |
|---|---|
| **Name** | Interactive Student Study Whiteboard |
| **Summary** | Provides students with a digital whiteboard within each classroom where they can write notes, draw diagrams, and organize study content visually. |
| **Priority** | Medium |
| **Actors** | Student, System |
| **Pre-condition(s)** | Student is logged in and enrolled in at least one classroom. |
| **Post-condition(s)** | Whiteboard content is saved and accessible to the student for future study sessions. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student navigates to the Whiteboard section within a classroom. | Displays the whiteboard tool with drawing and text options. |
| Student creates a new whiteboard or opens an existing one. | Loads the selected whiteboard canvas. |

| | |
|---|---|
| Student uses drawing, text, and shape tools to write notes. | Renders all inputs on the canvas in real time. |
| | Auto-saves whiteboard content continuously. |
| Student exports the whiteboard. | Generates and downloads the whiteboard as an image or PDF. |

**Alternate Course of Actions:**

| Actor Action | System Response |
|---|---|
| Auto-save fails due to a connection issue. | Alerts the student and stores changes locally until reconnection. |
| Student tries to export with an empty canvas. | Displays a warning and prevents empty export. |

# UC-10 – Pomodoro Timer for Study Management

| Identifier | UC-10 |
|---|---|
| **Name** | Pomodoro Timer for Study Management |
| **Summary** | Provides students with a built-in Pomodoro timer to manage focused study sessions and breaks, operating independently from AI services. |
| **Priority** | Low |
| **Actors** | Student, System |
| **Pre-condition(s)** | Student is logged in on the AiCo platform. |
| **Post-condition(s)** | Student completes a structured Pomodoro session with productivity records saved. |

**Typical Course of Actions:**

| Actor Action | System Response |
|---|---|
| Student opens the Pomodoro timer from their dashboard. | Displays the timer with configurable session and break durations. |
| Student sets session duration (default 25 min) and break (default 5 min). | Captures configuration settings. |
| Student clicks Start. | Begins countdown and displays remaining time. |
| Session timer reaches zero. | Alerts the student and automatically starts the break timer. |
| Break ends. | Prompts student to begin the next session; after 4 sessions, suggests a long break. |

**Alternate Course of Actions:**

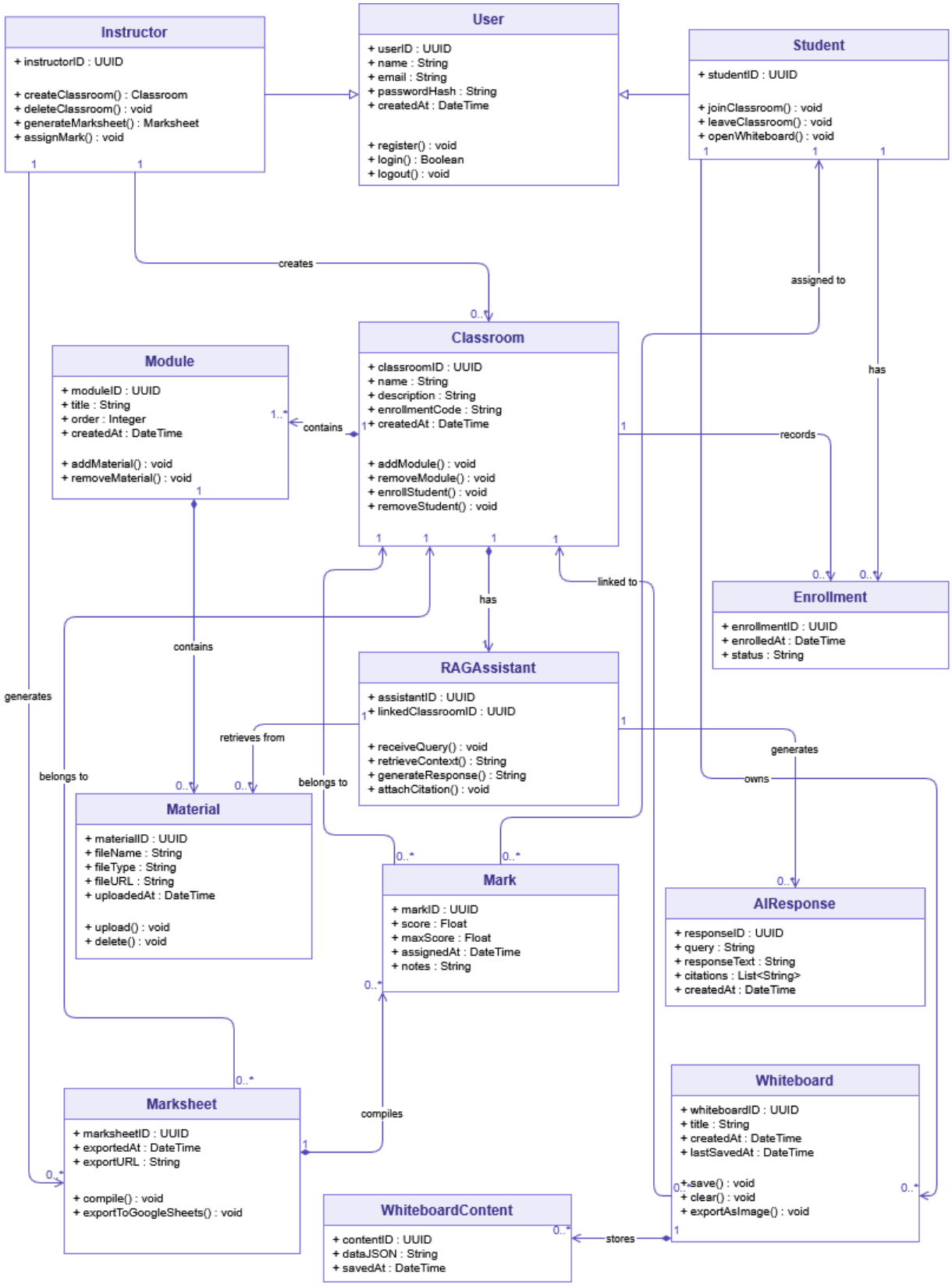
| Actor Action | System Response |
|---|---|
| Student pauses or resets the timer. | Pauses/resets without losing session count data. |

| | |
|---|---|
| Browser tab is closed mid-session. | Notifies student on return that the session was interrupted. |

# 3. Non functional requirements

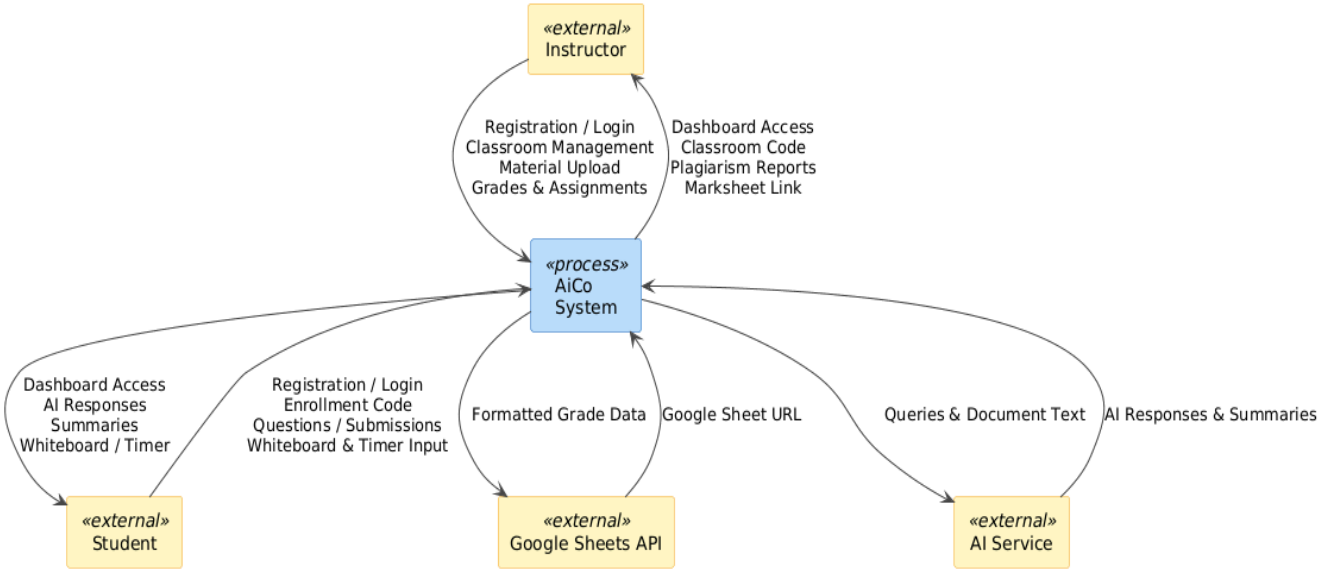
1. **Security:** All user passwords must be hashed using bcrypt , and all data transmission must occur over HTTPS to ensure secure authentication and account protection.
2. **Performance:** The system must respond to standard user interactions (login, enrollment, dashboard load) within 2 seconds under a concurrent load of up to 500 users.
3. **Scalability:** The platform must support at least 500 active classrooms and 10,000 registered users simultaneously without degradation in system performance.
4. **Reliability:** The system must maintain a minimum uptime of 99.5%, with uploaded learning materials stored using reliable cloud storage to prevent data loss.
5. **Privacy:** Student data, including queries, submissions, and grades, must be encrypted at rest and in transit, and must not be shared with or used by third-party services without explicit institutional consent.
6. **AI Response Accuracy:** The RAG-based AI assistant must strictly ground all responses in instructor-uploaded content and must attach source citations to every generated answer to prevent hallucination.
7. **Maintainability:** The system must be built using a modular architecture so that individual components (e.g., AI assistant, plagiarism engine, timer) can be updated or replaced without affecting other modules.
8. **Usability:** The platform must be fully functional on all modern browsers (Chrome, Firefox, Edge, Safari) and must be responsive and usable on varying screen sizes.
9. **Fault Tolerance:** The Pomodoro timer and whiteboard features must operate independently of AI services, ensuring their availability is unaffected by any AI service outage or slowdown.
10. **Accessibility and Interface Readability:** The user interface shall support accessibility features such as readable typography, proper color contrast, and dark mode to accommodate diverse user preferences and reduce visual fatigue during long study sessions.

# 4. Class diagram

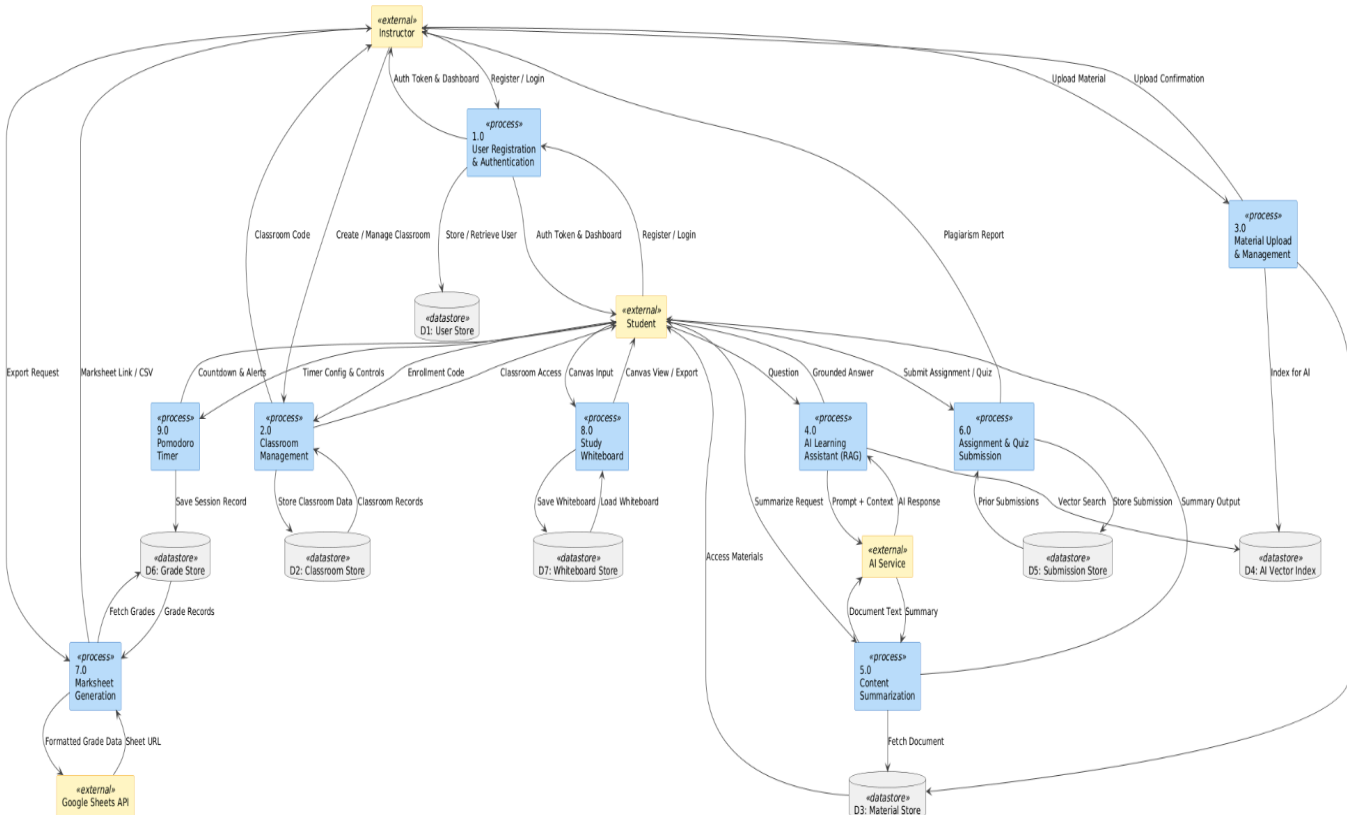


# 5. DFD diagrams

## 5.1 Level 0

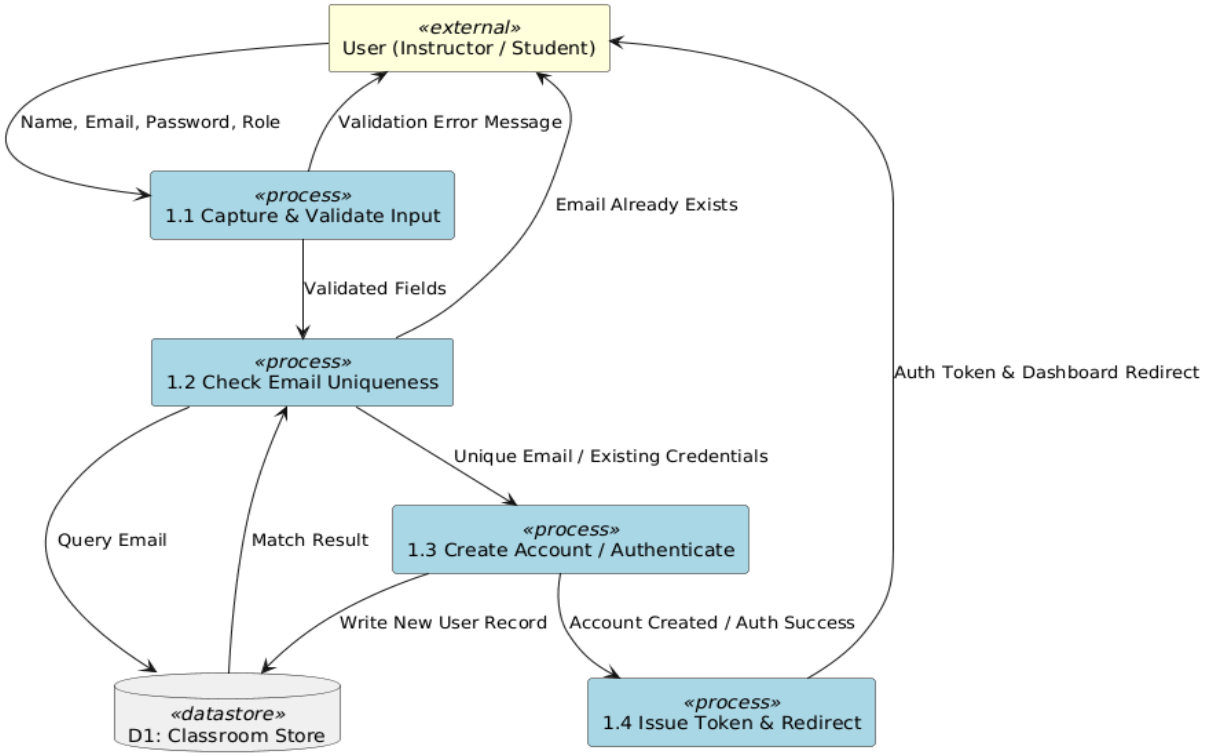



## 5.2 Level 1

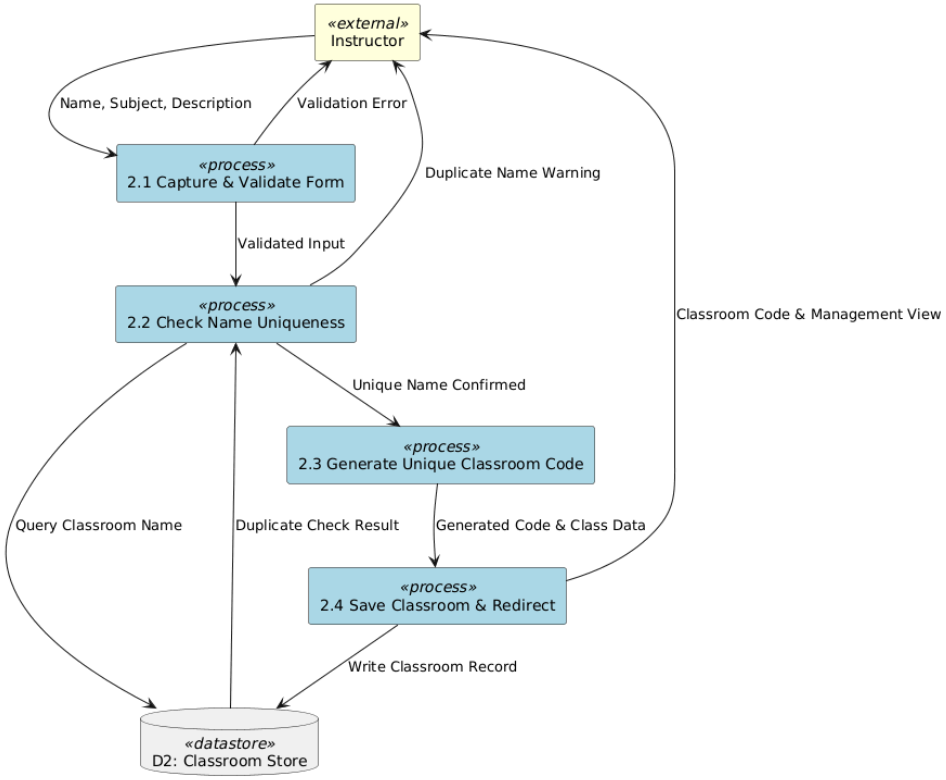


# 5.3 Level 2

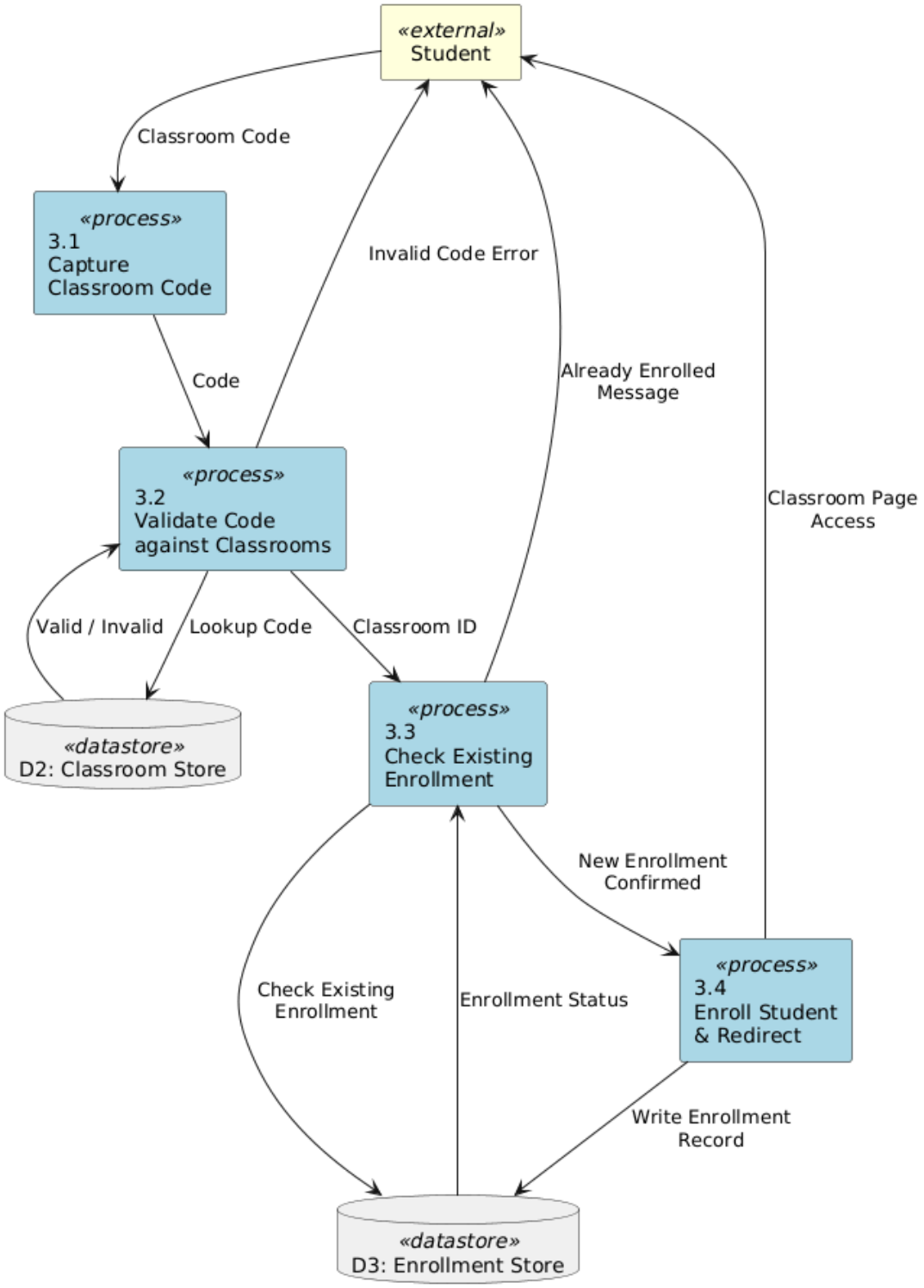
## 5.3.1 User authentication



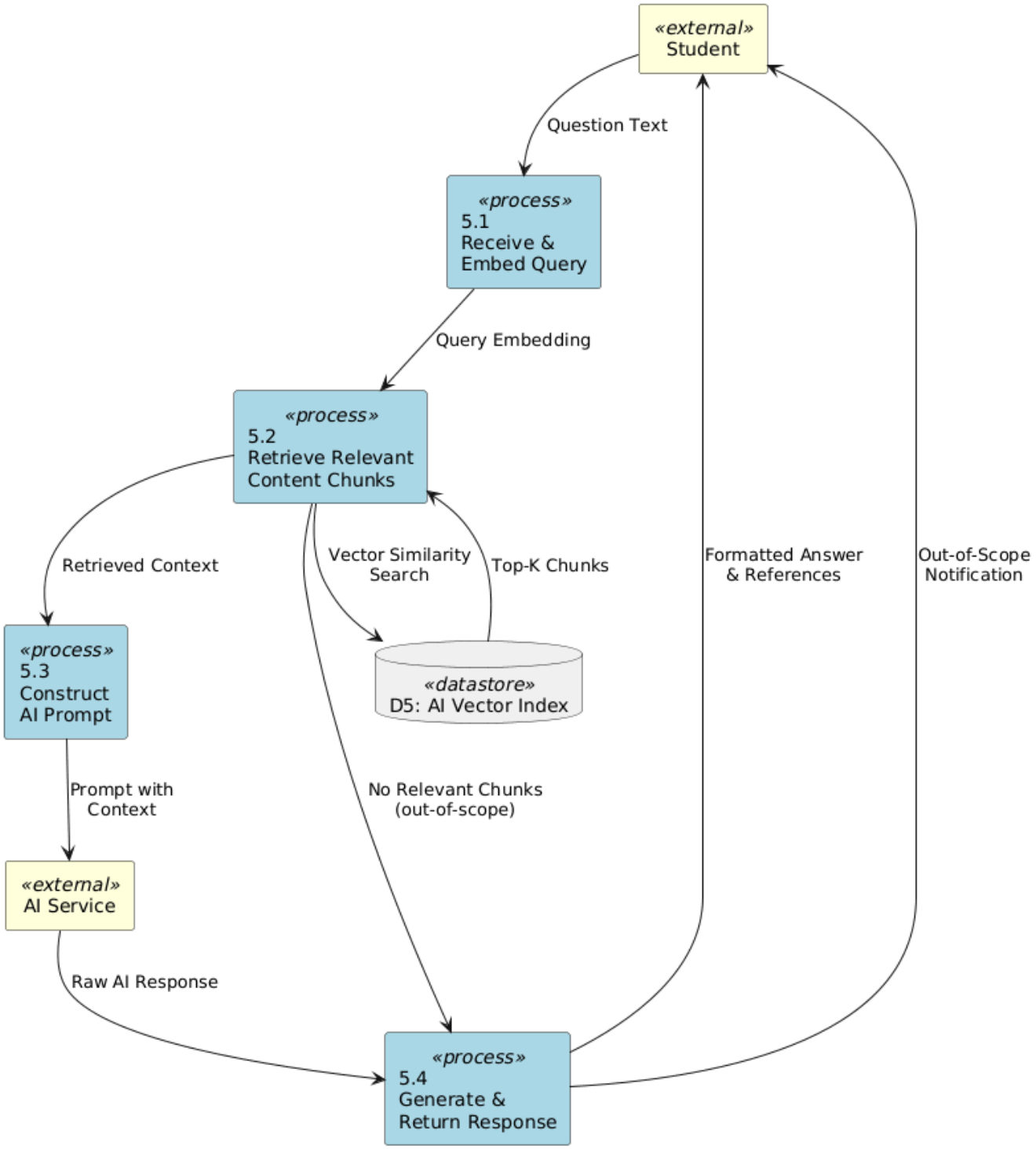

## 5.3.2 Classroom Management



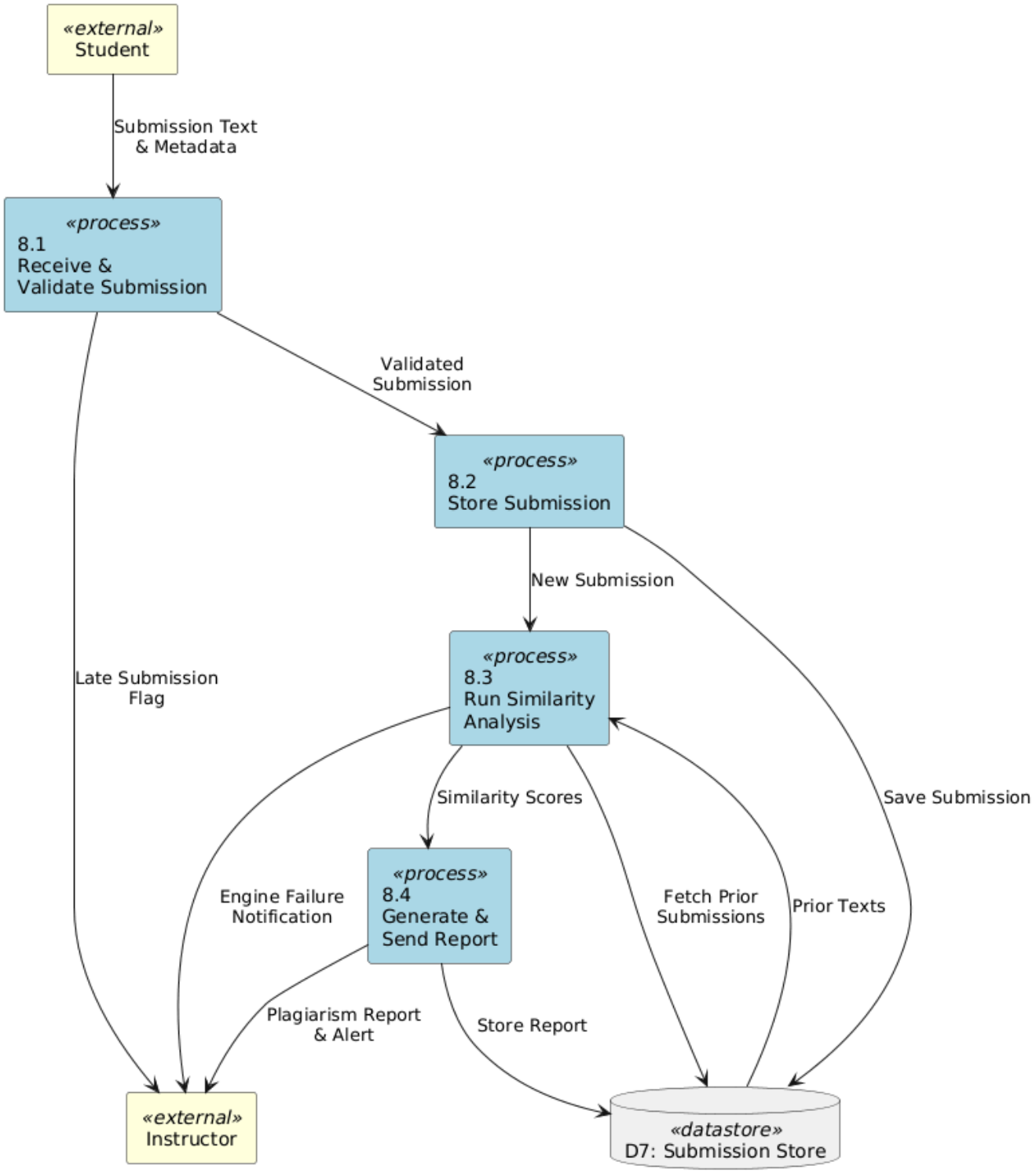
### 5.3.3 Student enrollments



### 5.3.4 Ai powered learning

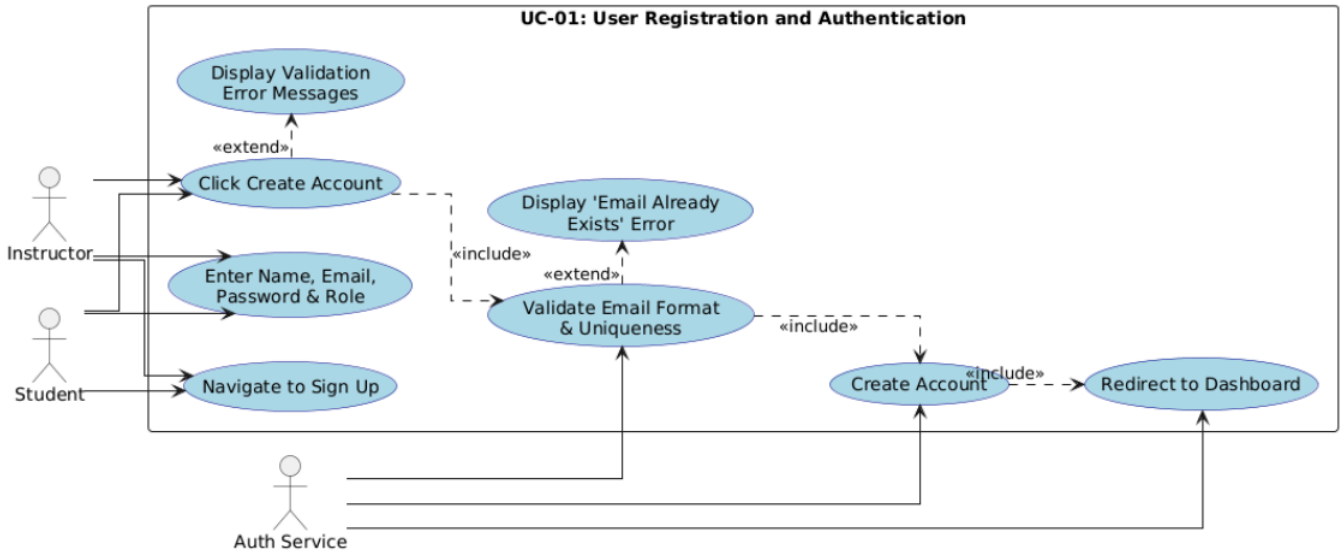


### 5.3.5 Quiz and Assignment submission with plagiarism checker

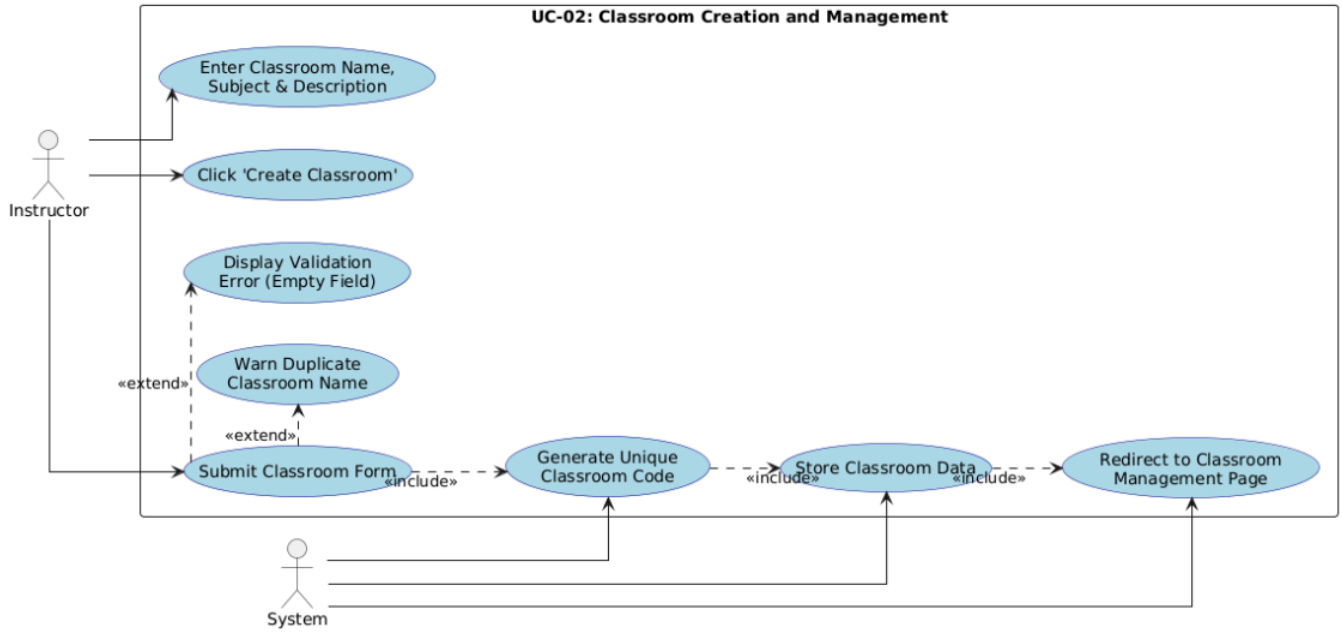


# 6.Use case diagrams

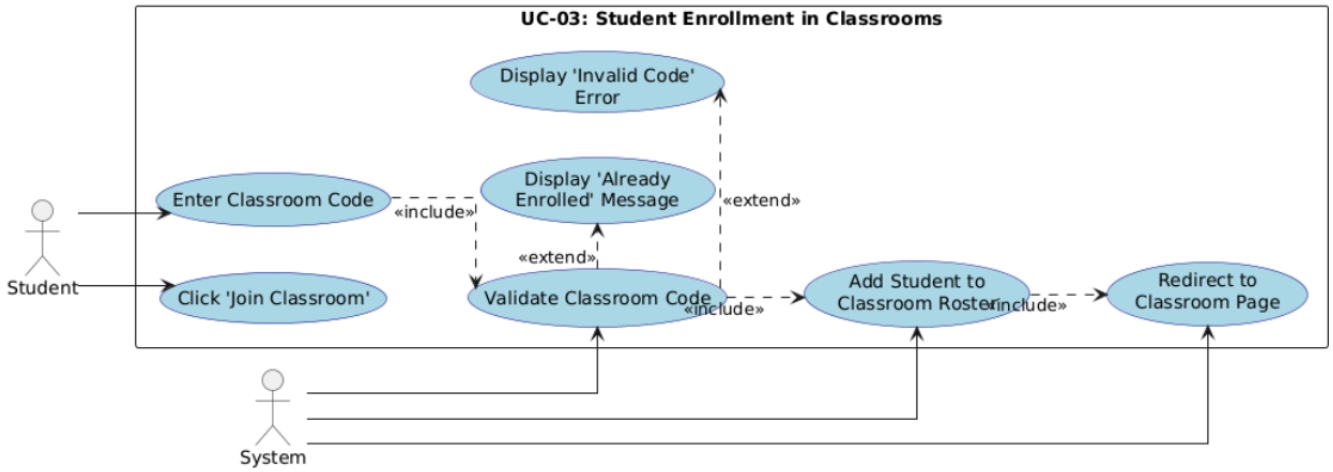
## UC-01 – User Registration and Authentication



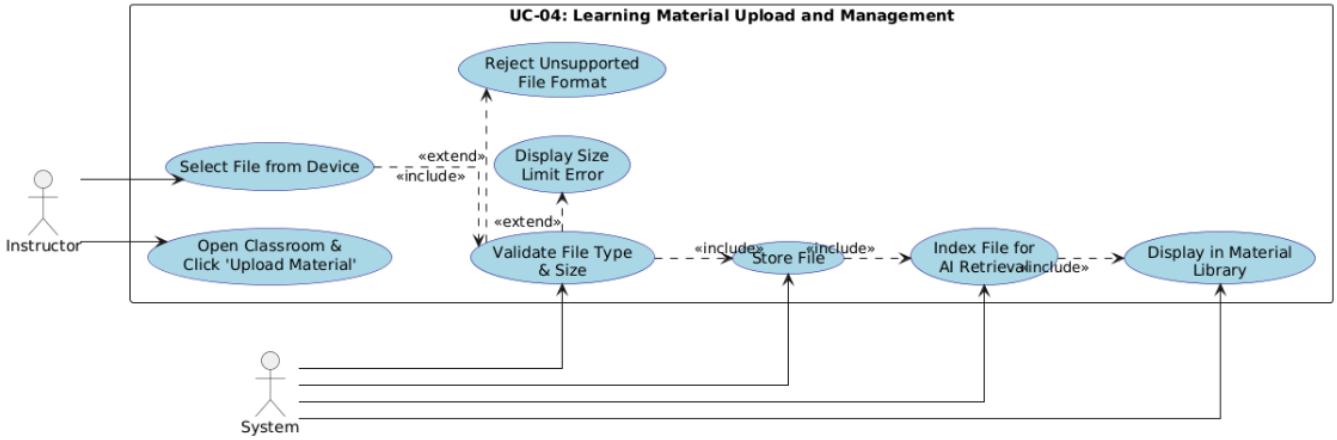

## UC-02 – Classroom Creation and Management



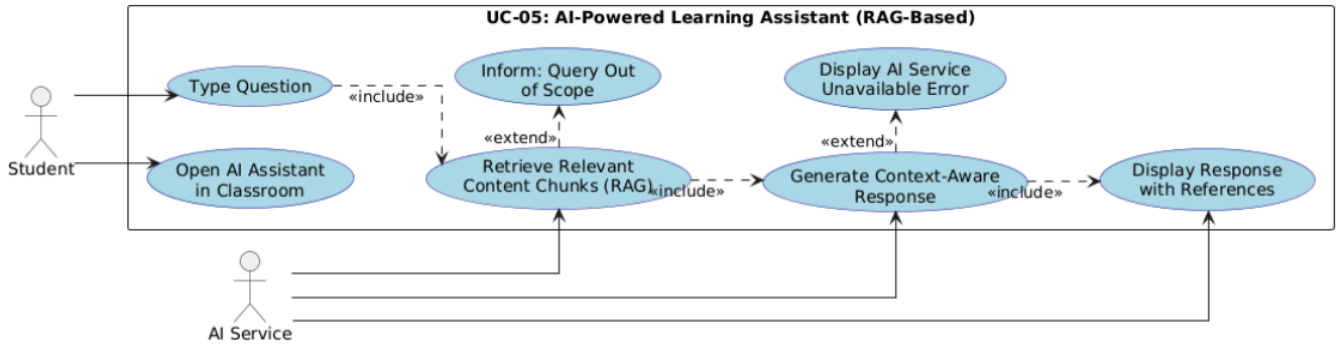
## UC-03 – Student Enrollment in Classrooms



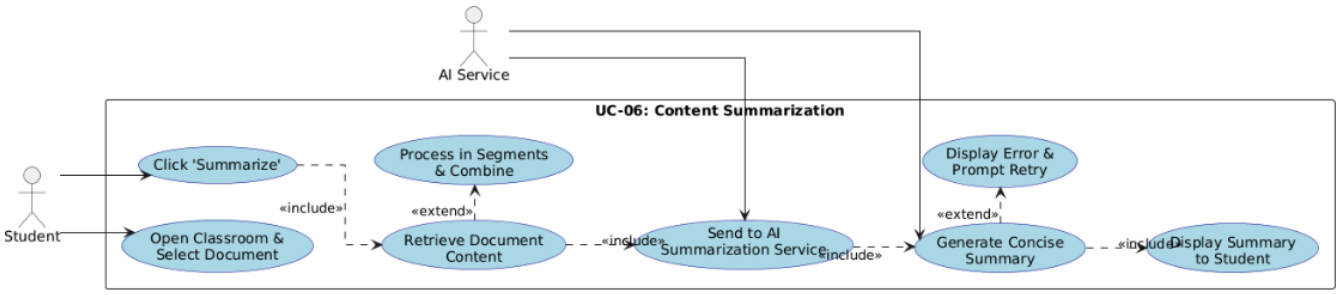

## UC-04 – Learning Material Upload and Management



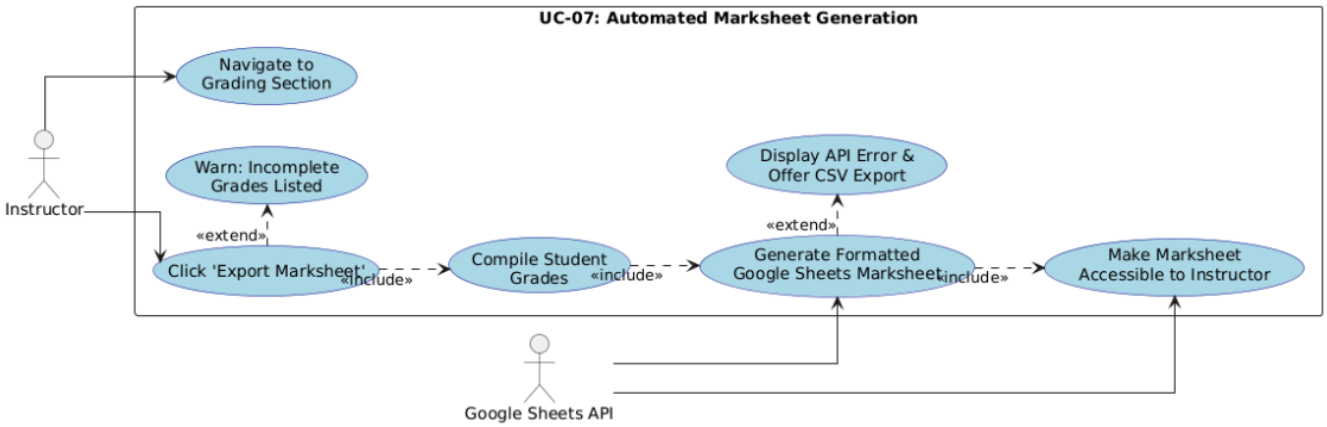

## UC-05 – AI-Powered Learning Assistant (RAG-Based)



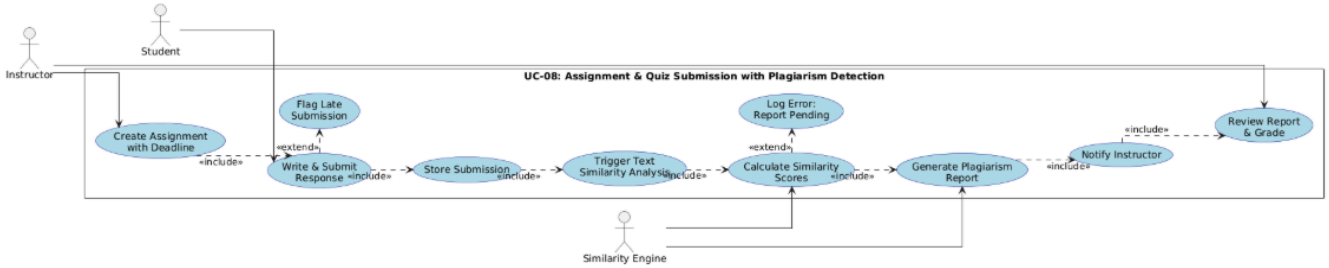
## UC-06 – Content Summarization



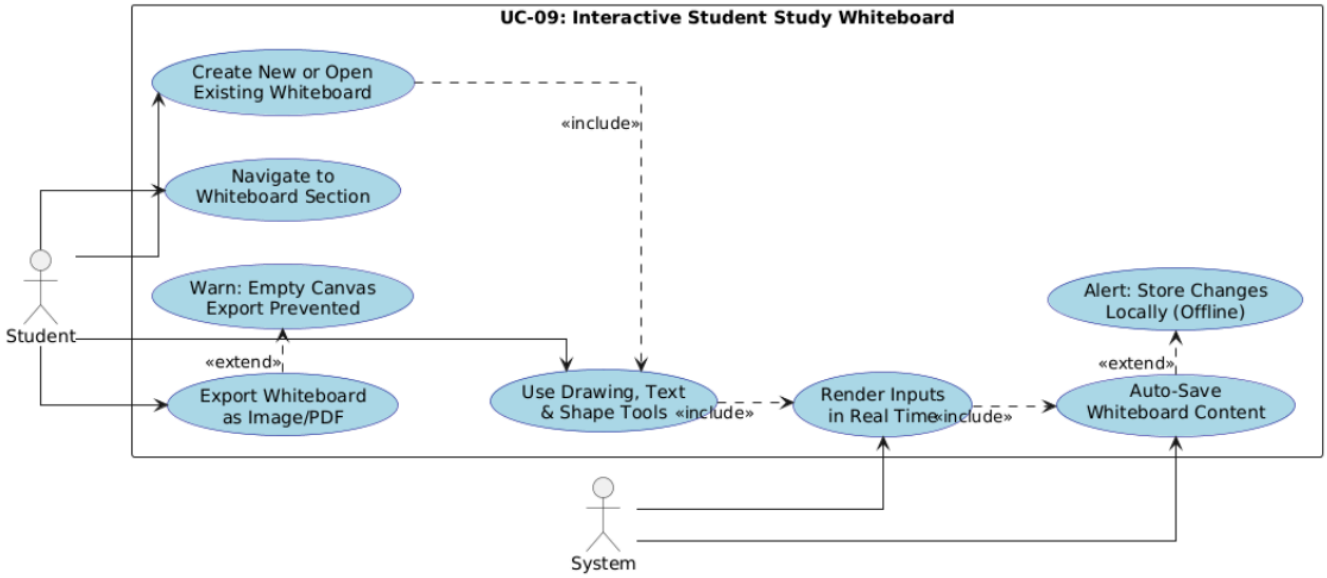

## UC-07 – Automated Marksheet Generation

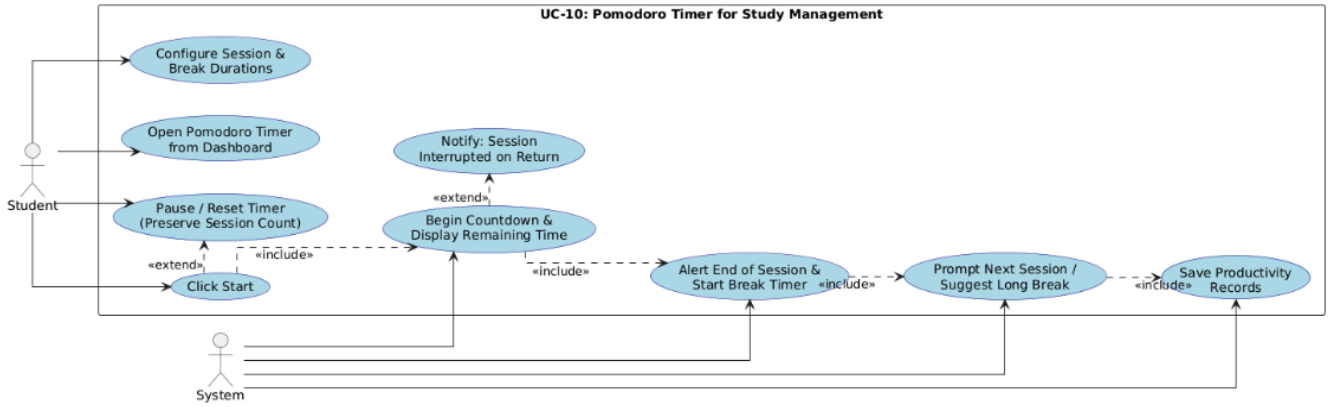



## UC-08 – Assignment and Quiz Submission with Plagiarism Detection



## UC-09 – Interactive Student Study Whiteboard




## UC-10 – Pomodoro Timer for Study Management